# Utkarsh Mathur

*Backend Software Engineer — Java / Spring Boot / Banking & Fintech Systems*

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## SUMMARY

Backend-focused Software Engineer with 2.5 years of experience building Java and Spring Boot applications in the banking and financial services domain. Contributed to core payment processing and SWIFT/ISO 20022 message migration work for a banking client. Recently built a full-stack loan origination system integrating an LLM-powered assistant using prompt engineering and retrieval-augmented generation (RAG). Microsoft- and Anthropic-certified in AI application development. Comfortable in Agile teams, CI/CD pipelines, and Dockerized deployments.

## EXPERIENCE

**Product Engineer** | Tata Consultancy Services (TCS)

*Gurugram, India* *August 2024 – Present*

- Developed and modified Core Java modules and XML-based configuration logic as part of a team executing an MT-to-MX (ISO 20022) migration for a Saudi Arabian banking client, supporting SWIFT-compliant message transformation.
- Built REST API components using Spring Boot, Spring MVC, and Spring Data JPA/Hibernate within a layered microservices architecture supporting core payment processing.
- Contributed to Spring Batch jobs that automated bulk transaction processing and reconciliation workflows handling 50,000+ daily financial records.
- Worked with IBM MQ and Kafka concepts to support asynchronous, high-volume financial messaging flows, and maintained SOAP/REST endpoints for banking data integrations.
- Tuned SQL queries and indexing strategies in Oracle DB, reducing query execution time by ~25%, and applied Singleton/Factory design patterns to improve code modularity.
- Used Jenkins CI/CD pipelines with Maven and Docker for build automation and deployment; debugged production issues using JVM-level tools, and collaborated within an Agile/Scrum team using JIRA and Git.

**Associate Software Developer** | Design2Occupancy LLP (D2O)

*Jaipur, India* *February 2024 – August 2024*

- Developed and maintained backend components for a corporate web portal using Java, Spring Boot, and JPA/Hibernate, including Oracle SQL schema design and query optimization.
- Built and tested modular REST APIs, writing unit tests to maintain code coverage and reliability.
- Participated in Agile ceremonies including stand-ups, sprint planning, and code reviews.

## PROJECT

**Loan Origination & KYC Onboarding System with AI Assistant**

- Built a full-stack loan origination and KYC onboarding system (Java 17, Spring Boot, PostgreSQL, Angular) covering customer onboarding, document metadata capture, and loan application status tracking.
- Integrated an AI assistant that answers applicant questions on required documents and loan policies using LLM API calls (OpenAI/Claude), grounded via a retrieval-augmented generation (RAG) pipeline with prompt engineering to reduce unsupported responses.
- Containerized the application with Docker Compose for local multiservice deployment of the backend and database.

## CERTIFICATIONS

**Microsoft Certified: Azure AI Apps and Agents Developer Associate** *Verify Credential*

*Microsoft | Issued Aug 2026 · Expires Aug 2027*

**Claude Certified Developer – Foundations** *Verify Credential*

*Anthropic | Issued Aug 2026 · Expires Aug 2027*

## EDUCATION

**Bachelor of Technology (B.Tech), Electrical Engineering**

*Jaipur Engineering College and Research Center | CGPA 8.57 | 2020–2024 | Jaipur, India*

## SKILLS

Languages & Databases: Java (8/17), SQL, XML, HTML5, CSS3, Oracle DB, PostgreSQL
Frontend, Backend, & Frameworks: Angular, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Batch, Spring Security, Servlets, JSP, JDBC.
AI & Generative AI: Prompt Engineering, LLM API Integration (OpenAI/Claude API), Retrieval-Augmented Generation (RAG), Azure AI Services
Banking, Messaging & Architecture: ISO 20022 (MT-MX Migration), SWIFT Compliance, Microservices, RESTful/SOAP APIs, IBM MQ, Kafka, Design Patterns
DevOps & Methodology: Jenkins, Docker, Maven, Git, JUnit, Mockito, Agile/Scrum, JIRA